

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Artificial Intelligence (01CT0616)	Aim: To obtain the classification of various classes using k-Nearest Neighbor approach	
Experiment No: 03	Date: 16/02/2026	Enrollment No: 92301733022

Aim: To obtain the classification of various classes using k-Nearest Neighbor approach

IDE: Google Colab

Theory:

Problems with classification models are resolved by this algorithm. In order to classify the data, the K-nearest neighbor, or K-NN, algorithm essentially draws an imaginary boundary. The program will attempt to forecast fresh data points as close to the boundary line as possible. Consequently, calmer curves of separation, which provide simpler models, are associated with greater k values. A smaller k number, on the other hand, tends to overfit the data and produce complex models. When examining the dataset, it is crucial to have the appropriate k-value to prevent both overfitting and underfitting.

The complete training dataset serves as the model representation for KNN. That is all there is to it. KNN does not require learning because its only model is to store the complete dataset. Complex data structures, such as k-d trees, can be used to store the data in efficient implementations, making it easier to find and match new patterns during prediction. You might want to carefully consider the consistency of your training data because the entire dataset is kept. Curating it, updating it frequently as new data becomes available, and eliminating inaccurate and outlier data could be wise.

The K-NN algorithm places the new case in the category that most closely resembles the existing categories based on the assumption that the new case and data are comparable to the available examples. The K-NN method classifies a new data point based on similarity after storing all of the relevant data. This implies that the K-NN technique makes it simple to classify newly discovered data into a well-suited category. Although the K-NN technique is mostly utilized for classification problems, it can also be used for regression.

K-NN is an algorithm that does not assume any specific structure of the data, making it non-parametric. It is referred to as a lazy learner algorithm because it does not immediately process the training set; instead, it simply retains the dataset and classifies new data when needed. During the training phase, the KNN algorithm merely saves the dataset, and when new data arrives, it assigns that data to a category similar to it.

Example: Suppose we have an image of a creature that looks like cat and dog, but we want to know whether it is a cat or dog. So, for this identification, we can use the KNN algorithm, as it works on a similarity measure. Our KNN model will find similar features to the cats and dog's images and based on the most similar features it will put it in either cat or dog category.

Methodology:

1. Load the basic libraries and packages
2. Load the dataset
3. Analyze the dataset
4. Pre-process the data

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5. Visualize the Data
6. Separate the feature and prediction value columns
7. Select the number K of the neighbors
8. Calculate the Euclidean distance of K number of neighbors
9. Take the K nearest neighbors as per the calculated Euclidean distance.
10. Among these k neighbors, count the number of data points in each category.
11. Assign the new data points to that category for which the number of the neighbor is maximum.
12. Our model is ready.

Pre Lab Exercise:

- a. What is K in KNN-approach?

Ans: When the algorithm generates a prediction, K indicates how many of the nearest d data points it uses.

- b. Which are the types of distance metrics?

Ans: Euclidean Distance, Manhattan Distance, Minkowski Distance, etc.

- c. What can be the criteria of selection of the value of K?

Ans: Prevent both underfitting and overfitting by applying cross-validation and selecting an odd value for K.

- d. What are the advantages of KNN algorithm?

Ans: It offers multi-class classification without making assumptions about the data distribution, handles non-linear patterns well, and doesn't require an explicit training phase.

Program (Code):

```
import time
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import load_digits
from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.preprocessing import StandardScaler
```

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```
from sklearn.pipeline import Pipeline
```

```
from sklearn.neighbors import KNeighborsClassifier
```

```
from sklearn.ensemble import RandomForestClassifier
```

```
from sklearn.neural_network import MLPClassifier
```

```
from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    confusion_matrix
)
```

```
digits = load_digits()
```

```
X = digits.data
```

```
y = digits.target
```

```
images = digits.images
```

```
plt.figure(figsize=(8, 4))
```

```
for i in range(10):
```

```
    plt.subplot(2, 5, i + 1)
```

```
    plt.imshow(images[i], cmap="gray")
```

```
    plt.title(f"{y[i]}")
```

```
    plt.axis("off")
```

```
plt.suptitle("Sample Digits")
```

```
plt.tight_layout()
```

```
plt.show()
```

```
X_train, X_test, y_train, y_test = train_test_split(
```

```
    X, y, test_size=0.2, random_state=42)
```

```
knn_pipeline = Pipeline([
```

```
    ("scaler", StandardScaler()),
```

```
    ("knn", KNeighborsClassifier())
```

```
])
```

```
knn_param_grid = {
```

```
    "knn__n_neighbors": [3, 5, 7, 9, 13, 15, 20]}
```

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```
rf_pipeline = Pipeline([
    ("rf", RandomForestClassifier(random_state=42))
])
```

```
rf_param_grid = {
    "rf__n_estimators": [25, 50, 100, 200, 500],
    "rf__max_depth": [None, 10, 20]}
```

```
mlp_pipeline = Pipeline([
    ("scaler", StandardScaler()),
    ("mlp", MLPClassifier(
        max_iter=1000,
        random_state=42,
        early_stopping=True
    ))
])
```

```
mlp_param_grid = {
    "mlp__hidden_layer_sizes": [(64,), (128,), (64, 64)],
    "mlp__activation": ["relu", "tanh"],
    #"mlp__alpha": [0.0001, 0.001],
    "mlp__learning_rate_init": [0.001, 0.01, 1.0, 5.0, 10.0]
}
```

#Training the model

```
def train_with_timing(pipeline, param_grid):
    start = time.time()
    grid = GridSearchCV(
        pipeline,
        param_grid,
        cv=5,
        scoring="f1_weighted",
        n_jobs=-1
    )
    grid.fit(X_train, y_train)
    train_time = time.time() - start
    return grid, train_time
```

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```
knn_grid, knn_train_time = train_with_timing(knn_pipeline, knn_param_grid)
rf_grid, rf_train_time = train_with_timing(rf_pipeline, rf_param_grid)
mlp_grid, mlp_train_time = train_with_timing(mlp_pipeline, mlp_param_grid)
```

#Inferencing the model

```
def inference_time(model):
    start = time.time()
    model.predict(X_test)
    return time.time() - start
```

```
knn_test_time = inference_time(knn_grid.best_estimator_)
rf_test_time = inference_time(rf_grid.best_estimator_)
mlp_test_time = inference_time(mlp_grid.best_estimator_)
```

#Evaluation

```
def evaluate_model(model):
    y_pred = model.predict(X_test)
    return {
        "Accuracy": accuracy_score(y_test, y_pred),
        "Precision": precision_score(y_test, y_pred, average="weighted"),
        "Recall": recall_score(y_test, y_pred, average="weighted"),
        "F1-score": f1_score(y_test, y_pred, average="weighted")
    }
```

```
knn_metrics = evaluate_model(knn_grid.best_estimator_)
rf_metrics = evaluate_model(rf_grid.best_estimator_)
mlp_metrics = evaluate_model(mlp_grid.best_estimator_)
```

#Result with comparison

```
results_df = pd.DataFrame(
    [knn_metrics, rf_metrics, mlp_metrics],
    index=["KNN", "Random Forest", "MLP"]
)
```

```
results_df["Training Time (s)"] = [
    knn_train_time, rf_train_time, mlp_train_time
]
```

```
results_df["Inference Time (s)"] = [
```

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```

knn_test_time, rf_test_time, mlp_test_time
]

print(results_df)

results_df.iloc[:, :4].plot(kind="bar", figsize=(10, 6))
plt.title("Model Performance Comparison (Digits)")
plt.ylabel("Score")
plt.ylim(0, 1)
plt.xticks(rotation=0)
plt.grid(axis="y")
plt.show()

models = {
    "KNN": knn_grid.best_estimator_,
    "Random Forest": rf_grid.best_estimator_,
    "MLP": mlp_grid.best_estimator_
}

fig, axes = plt.subplots(1, 3, figsize=(18, 5))

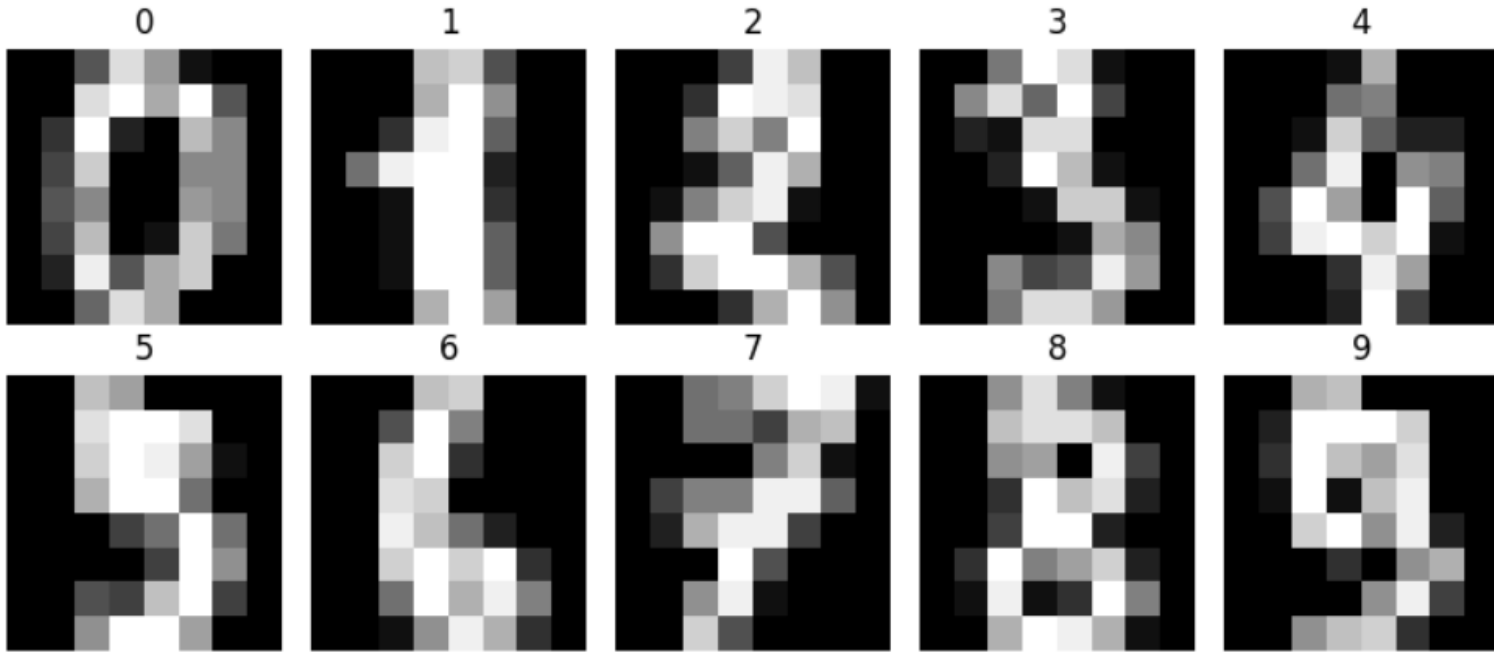
for ax, (name, model) in zip(axes, models.items()):
    cm = confusion_matrix(y_test, model.predict(X_test))
    sns.heatmap(cm, annot=True, fmt="d", cmap="Blues", ax=ax)
    ax.set_title(name)
    ax.set_xlabel("Predicted")
    ax.set_ylabel("Actual")

plt.tight_layout()
plt.show()

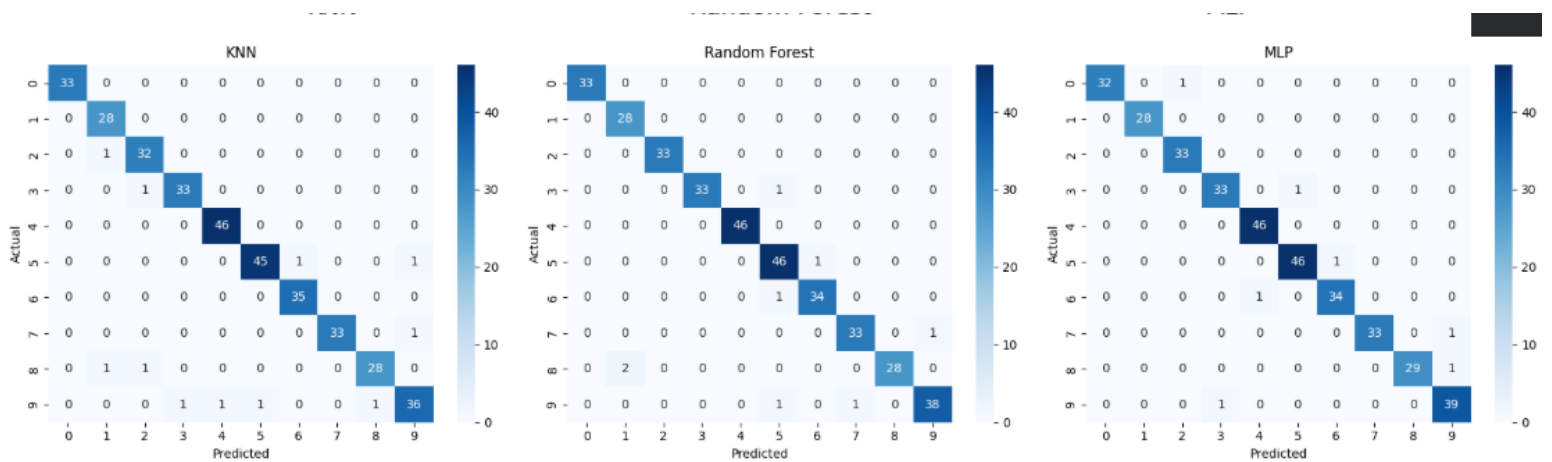
```

Results:

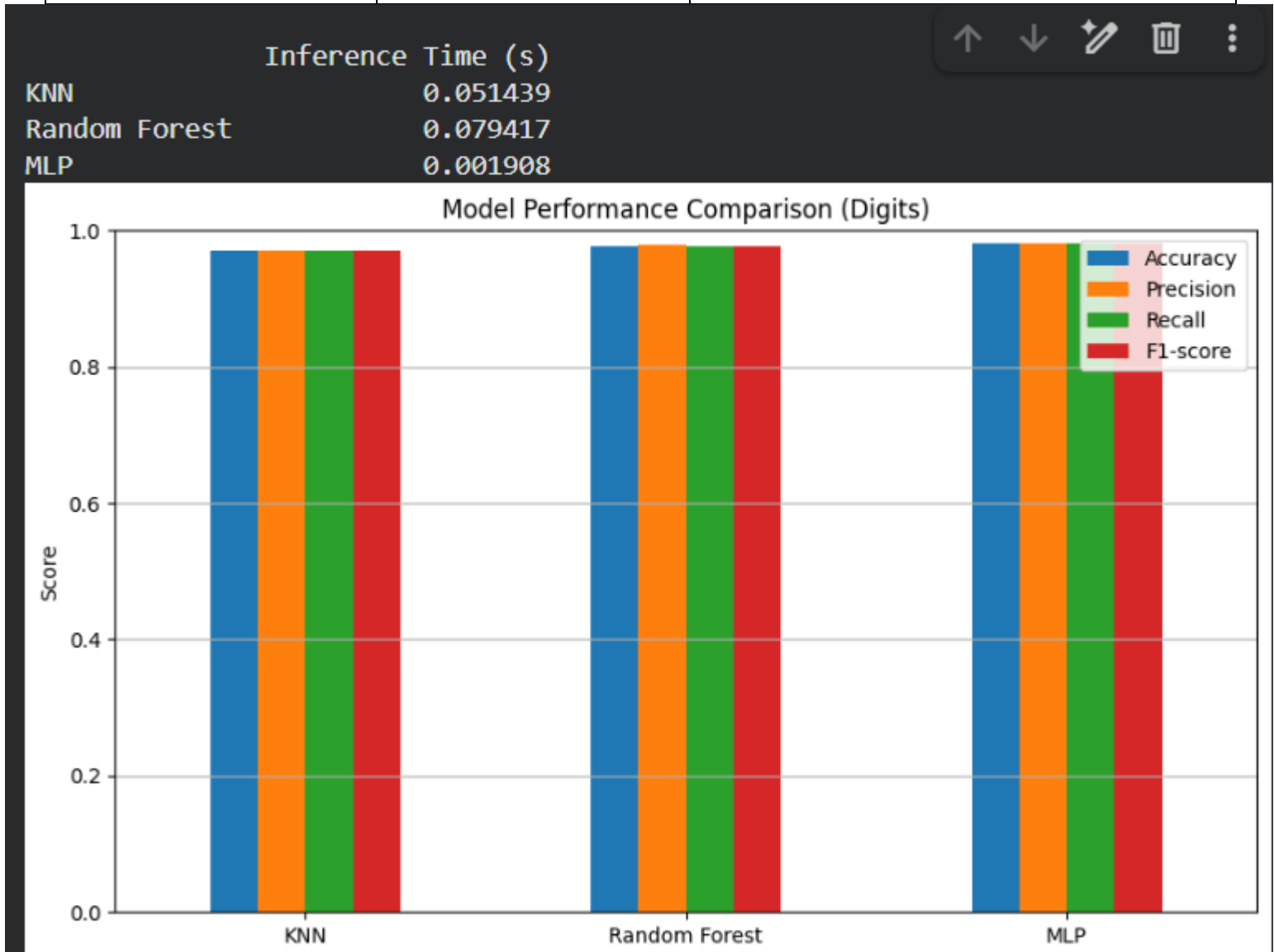
Sample Digits



	Accuracy	Precision	Recall	F1-score	Training Time (s) \
KNN	0.969444	0.969666	0.969444	0.969287	7.745977
Random Forest	0.977778	0.978417	0.977778	0.977820	61.980994
MLP	0.980556	0.980832	0.980556	0.980577	39.132682



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Observation and Result Analysis:

a. Nature of the dataset

Ans: The dataset, which consists of numerical pixel features that represent handwritten numbers (0–9), is a supervised, multi-class classification dataset.

b. During Training Process

Ans: The dataset is utilized as a supervised labeled dataset during training, where models use training data to identify patterns between pixel features and digit labels and cross-validate their performance.

c. After the training Process

Ans: The dataset is utilized as test data after training in order to assess the performance of the trained model

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by generating predictions and computing metrics such as accuracy, precision, recall, and F1-score.

d. Observation over the Learning Curve

Ans: The learning curve illustrates how training enhances model performance. It assists in determining when the model reaches stable and optimal performance as well as overfitting and underfitting.

Post Lab Exercise:

a. Is KNN useful in regression-based problems?

Ans: Yes, KNN is useful in regression problems. It predicts output by taking the average of values of K nearest neighbors instead of majority voting.

b. Why KNN is a non-parametric algorithm?

Ans: KNN is a non-parametric algorithm because it does not assume any fixed data distribution or mathematical model and makes predictions directly using stored training data.

c. What are the assumptions of KNN approach?

Ans: KNN assumes that similar data points are located close to each other, distance measures correctly represent similarity, data is properly scaled, and sufficient training data is available for accurate prediction.

d. How can the KNN approach make the prediction of the unseen dataset?

Ans: In order to predict unknown data, KNN computes the distance from each training point, chooses the K nearest neighbors, and then uses either majority voting (classification) or averaging (regression) to provide the final prediction.

e. Why is KNN called a Lazy Learner?

Ans: The reason KNN is referred to as a lazy learner is that it does not construct a model during training and only carries out all computations when predicting fresh data.

f. Why should we not use KNN for large dataset?

Ans: Because KNN must compute the distance from every training data point for every new input, it demands a lot of memory, high computation time, and slow prediction, making it unsuitable for huge datasets.